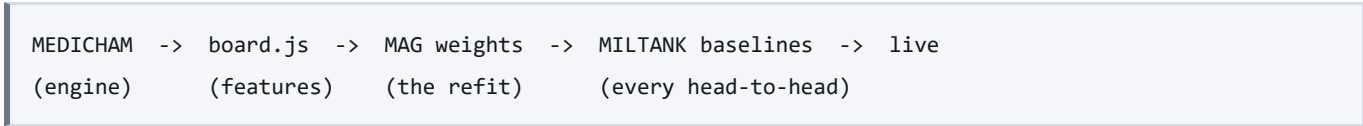


ABRA — Project Summary

Version 7.0.0 · 2026-09-20 · Will Hooper

7.0.0 — THE MEDICHAM QUARANTINE GATE IS OPEN. THE SIMULATOR AGREES WITH SHOWDOWN ON EVERY INSTRUMENT THAT MEASURES IT, INCLUDING A HELD-OUT SAMPLE THE GATE DOES NOT READ. EVERY MODEL DOWNSTREAM OF IT BECOMES RE-RUNNABLE, NONE HAS BEEN RE-RUN, AND SO NOT ONE OF THEIR FIGURES APPEARS IN THIS DOCUMENT.

ABRA is the Automated Battle Replay Analyzer. It ingests public Champions Reg M-B replays from Pokémon Showdown, models the ladder metagame, and feeds the model CHOMP reads. Underneath that sit a simulator (MEDICHAM), a board evaluation, a fitted action model (MAG) and a search player (MILTANK). The dependency runs one way, which is why a wrong simulator was never an engine-only problem:



Everything right of the first arrow has been withheld since 2026-08-08, because the box on the left was known incorrect. That is what changed today, and it changed for the left-hand box only.

1. Every component, and the state it is in

component	what it is	state today
MEDICHAM — the simulator	plays Champions Reg M-B locally so a position can be rolled out	GATE OPEN. All ten clauses of <code>node engine/quarantine.js</code> pass on release <code>0d7b1d9db6d1</code>
The Showdown differential	plays the same game on both engines and compares the board at every turn boundary	LIVE and clean — see §2. It measures MEDICHAM, so it was never quarantined
The damage differential	stages damage stage by stage against the authority	LIVE and clean at every interior index — see §2
The deliberate roster	one staged scenario per legal item, ability and move	LIVE and clean on all three stages — see §2
The mechanics census and the staged harness	probes a mechanic inside a real game and reads the verdict off Showdown	LIVE and clean — see §2
<code>board.js</code> — the features	turns a position into the numbers every model scores	moves with the simulator; nothing in it is published as a figure
MAG — the fitted action model	scores one candidate action	WITHHELD. A re-run is owed, and a REFIT before it — see §3 and §4
The joint layer	scores both of my slots together on one turn	WITHHELD. Same fit, same refit
MILTANK — the search player	chooses leads, brings, mega timing and post-KO	WITHHELD. Every baseline it holds was taken through a rollout
GARY — the opponent inside the search	models what the other side does while MILTANK searches	WITHHELD. Reads a rollout
PORYGON2 / DODUO / SLOWKING	position value, one-turn joint policy, the bring	WITHHELD. Each reads a rollout

component	what it is	state today
Leaf calibration — MEASURE's one number	does the leaf's stated win probability match the frequency actually observed	NOT MEASURED on this engine. No replication verdict, no bucket share, no calibration gap, no sample size — and no direction may be read into the absence
The store and the ingest	every replay kept raw, forever, with rating and bot tags	LIVE, six-hourly — see §2
META-USAGE — the model CHOMP reads	usage, threats and bring priors off the store	LIVE — see §2
The next-regulation collector	pulls Reg M-C games hourly so the corpus exists before the work does	COLLECTING ONLY. Nothing about M-C is simulated, and nothing here is a claim about it
ABRA WORLD — the site	renders what the artifacts say, and never authors a number	PAUSED. Its bundles were built 2026-08-25 and do not describe today's gate; <code>node engine/status.js</code> calls the drift by name
The honesty machinery	<code>status.js</code> , <code>provenance.js</code> , <code>quarantine.js</code> , the docs gate	LIVE, and two of its own ratchets are red — see §4

Nothing in the WITHHELD rows is a judgement about quality. It is the rule this project has run since 2026-08-08: a figure taken through a simulator that has been repaired since is not a bad figure, it is an unrepeated one. The gate opening makes those artifacts **re-runnable, not true.**

2. What is measured, and what it reads

All of it on engine release `0d7b1d9db6d1` , with the census pinned and the team pool pinned to `data/team-pool-frozen` . The flags are part of the sample definition, so they are named in the row.

instrument	reading	read from
held-out wide sample, <code>--games 12000</code>	7,182 games, 0 board-material	recorded in CHANGELOG <code>6.83.0</code> ; readout below
gate lattice A, <code>--games 1200</code>	961 games, 0 board-material, 0 undeclared narration; all 10,716 compared turn boundaries identical, <code>turns_cap 50</code> , pool digest <code>0d103fb9fa87</code> off 8,778 teams with 1,968 picked	<code>data/game-differential.json</code>
gate lattice B, <code>--games 1350</code>	1,069 games, 0 board-material, 0 undeclared narration	<code>data/game-differential.g1350.json</code>
gate lattice C, <code>--games 1950</code>	1,497 games, 0 board-material, 0 undeclared narration	<code>data/game-differential.g1950.json</code>
the damage differential	6,000 compared, 0 disagreed at the midpoint and at every interior index, seed 20260804	<code>data/engine-diff.json</code>
deliberate roster, items	148 of 148 in scope tested, 0 differ, 0 did-not-fire	<code>data/roster.items.json</code>
deliberate roster, abilities	196 of 200 in scope tested, 0 differ, 0 did-not-fire	<code>data/roster.abilities.json</code>
deliberate roster, moves	496 of 497 in scope tested, 0 differ, 0 did-not-fire	<code>data/roster.moves.json</code>
the mechanics census	1,002 probed / 1,002 live / 0 missing	<code>data/mechanics-census.json</code>
the staged harness	4,632 games played, 0 threw; 497 moves, 200 abilities and 148 items in scope, 0 diverging	<code>data/all-mechanics-fire.json</code>
the store	94,360 games recorded, 33,539 usable (35.5%), 28,681 teams	<code>data/live.js</code>
META-USAGE	94,360 collected, a funnel ending at 33,539 clean games, 67,078 sampled teams	<code>data/meta-usage.json</code>

instrument	reading	read from
the Reg M-C corpus	32,997 singles games and 23,616 bo3 games held	data/next-regulation.json

The held-out sample is drawn by a stride the gate does not use, so it is a different lattice rather than a larger one. Its artifact is not a top-level `data/*.json` file, so its readout is quoted here rather than cited:

```
$ node -e "const j=require('./data/verification/game-differential.g12000.json');const s=j.state;console.log(
0d7b1d9db6d1  games 7182  board_never_diverged 7182  protocol_diverged 75  protocol_diverged_board_nev
```

Read the second half of that line as carefully as the first. Board-material is zero on the wide sample. Protocol divergence is not: each of those games parts a line of commentary somewhere and no board anywhere. The narration gate is measured on the three lattices, where it reads zero undeclared; on the wide sample narration is reported, and it is not zero.

3. What is withheld, and what would end it

The list is derived, never typed — `node engine/quarantine.js` walks the require graph and names every artifact downstream of the simulator, and `node engine/status.js` prints what re-runs each one.

withheld	what re-runs it
leaf calibration	<code>node engine/backtest_winrate.js</code>
R1 leaf accuracy	<code>node engine/rollout_r1_artifact.js</code>
R2 leaf cost	<code>node engine/rollout_r2.js</code>
R3 divergence	<code>node engine/rollout_r3.js</code>
R4 head-to-head	<code>node engine/rollout_r4.js</code>
engine correctness → leaf	<code>node engine/leaf_engine_contrast.js</code>
click censoring	<code>node engine/click_census.js</code>
the MAG and joint weights	a REFIT and not a restamp: <code>node engine/fit_policy.js</code> , then <code>node engine/fit_joint.js</code>
every model report that reads a rollout — MAG, MILTANK, GARY, PORYGON2, DODUO, SLOWKING	each has its own generator; <code>node engine/status.js</code> names it beside the withheld figure

The re-run is owed and none of it is done. Will's instruction of 2026-09-09 stands: leave them until MAG and MILTANK are reworked. So this document quotes no model figure at all — not a head-to-head, not an exploitability share, not a leaf accuracy, not a calibration gap.

4. The honest column

The zero is a statement about what was measured. Four lattices and a staged lab, on one release, over a frozen pool of real ladder games. It is not a proof of equivalence, and a lattice that contains no divergence is not the same claim as an engine that cannot produce one — that lesson was bought on 2026-09-12, when one sample read zero and two wider ones did not.

Illusion is the one declared exclusion. The differential drops teams carrying a legal Illusion body rather than modelling the disguise, and the gate excuses those rows by name. Derived from the format and from the frozen pool rather than recalled:

```
legal carriers, from Dex.forFormat('gen9championsvgc2026regmb'): zoroark, zoroarkhisui
data/team-pool-frozen/games.bo3.jsonl    13,214 games    26,428 sides (one side = one player's team in one
sheets carrying a legal Illusion body    452 of 26,428 = 1.71%    of those 452, brought one: 229
```

An earlier statement of this exclusion divided a SHEET count by a GAME count, which inflated the rate about twofold. The readout above is the re-derivation, and the inflated pair appears nowhere in this document — deleted rather than footnoted.

A clean roster stage is not a fully tested one, and the gaps are declared. Three ability rows pass as ANNOUNCEMENT-ONLY on a recorded receipt — a restore knob and a probe each — and one is deferred by the owner (`data/roster.abilities.json`); the moves stage carries one deferred row as well (`data/roster.moves.json`). The staged harness shelves its diverging rows by the owner's decision, and every one of them is staged on a legal Illusion carrier — `data/all-mechanics-fire.json` `summary.moves.shelved_by_owner_diverging` 2 and `summary.abilities.shelved_by_owner_diverging` 1. A shelved row is still staged and still played; it does not vote.

Closed team sheets and bo1 are out of scope this release. Every figure in §2 describes open-sheet play. Nothing here says how the engine, or anything fitted on it, behaves when the sheet is hidden.

Seven live `lastMove` readings are an unregistered class. The authority keeps the CALLING move where this engine keeps the called one. Reported, not fixed, and carrying no roadmap row.

Two of the ten gate clauses cannot be computed without a local Showdown checkout. On a machine with no `SHOWDOWN_PATH` the board-leaf clause and the mechanics clause read CANNOT-ANSWER — which fails, and never reads as a pass. The OPEN verdict above is a reading taken where the authority was present.

The release these figures name is not in the repository. `data/releases/` is ignored by git and `0d7b1d9db6d1` was not force-added, so `node engine/provenance.js` marks every artifact citing it PUBLISHED FIGURE ON AN UNTRACKED RELEASE . The manifest digests still say what was frozen; from a fresh clone the evidence chain ends at that string rather than at bytes somebody can re-open.

MAG's fitted vector is red on its own check, and it is the first thing owed.

`engine/feature_fixture.js` fires two gates against the shipped weights: the fixture itself changed, and **the damage table those weights were fitted against has been regenerated**, so the table digest stamped beside the fit is not the digest of the table on disk. A restamp answers the fixture gate and silences the table gate in the same stroke, which is why the order matters — settle the table verdict, then refit. `node engine/status.js` prints it as REFIT OWED on every run.

Two of this project's own ratchets are red. `node engine/provenance.js` exits non-zero: six `_diag*` artifacts ship without recording what content they read. And the bundles under `web/` publish a gate state the live gate no longer has.

The interaction matrix has not been re-run on this engine. Its last run predates the release above by weeks, so no agreement figure from it appears here.

5. Why this is a MAJOR, and how the old numbers link to the new

A MAJOR here means the BASIS moved — the question the numbers answer, rather than their values. Three things a reader can no longer be told.

1. **That MEDICHAM is not correct.** That sentence has stood under every model figure in this project since 3.79.0, and it is why they were withheld. It is now false, and the withholding that rested on it ends at

once.

2. **That a whole-game figure from this project describes an engine whose defects are unknown.** The instruments in §2 bound the engine's disagreement with the authority at zero on every board they compare; what is left is narration, and it is reported as narration.
3. **That the withheld models are merely out of date.** They are not comparable to anything measured today: they were taken through a simulator repaired many times since, so old and new cannot be linked by arithmetic. They have to be re-run, which is why no number from them is restated here.

The back-cast, so the two series can be linked. The held-out board partings across 2026-09-20 ran **34** → **15** → **9** → **1** → **0**, recorded in CHANGELOG 6.83.0, each step on its own engine release with the pool and the census pinned. That sequence is the bridge between the last published series and this one.

Every whole-game count this document carried at 6.0.0 and earlier is DELETED, not captioned. Those readings belong to other engine bytes, other turn caps and other sample definitions; a caption beside them is the failure this project has already paid for twice. What replaces them is §2 — one release, one pool, and the flags written beside every row.

6. Where the current state is read

State is printed, never typed. Nothing in this document outranks what these print today.

```
node engine/status.js           # every figure, with the artifact it came from; NOT DERIVED where none s
node engine/quarantine.js       # the gate, clause by clause, and every artifact downstream of the simul
node engine/open_work.js        # every unclosed register row, and every defect a live instrument measur
node engine/docs_scan.js --owed # what the documents still owe the next major
```

docs/RUNNING-NOTES.md is the log between majors, and this document is the fold-in of those rows at 7.0.0. The division ledgers — docs/{ENGINE,MEASURE,SEARCH,OPS,WEB}.md — carry the working detail, and docs/_reports/ carries the dated accounts. None of the three is state.